CRBCL Technical Implementation, Live Verification & Evidence Appendix

Chief Red Bear Children's Lodge (CRBCL)

Audience: Technical Leads, Systems Auditors, Privacy Officers, Software Reviewers

Repository Baseline: Commit 6d28568 (feat(admin): add secure user Excel export) | Working Tree: Clean

Date: September 28, 2026 | Status: Forensic Technical Evidence & Verification Register

> ⚠ **Live Website Verification Notice & Operational Assessment Boundary**
>
> CRBCL has reported that not all capabilities represented in the current source implementation are functioning as expected on the live website. Accordingly, this report distinguishes source-code implementation from confirmed live operation. No capability is represented as production-operational solely because corresponding source code or automated tests exist. Reported live issues require authenticated end-to-end UAT and remediation where necessary.
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> • **"Technical readiness supports controlled UAT using synthetic and de-identified data."**
> • **"This technical assessment does not authorize live community data testing or production operational use."**
> • **"Technical controls have been assessed for alignment support; statutory applicability and legal compliance require CRBCL privacy/legal review."**

1. Repository Provenance & Baseline State

- **Current Repository HEAD:** Commit `6d28568` (`feat(admin): add secure user Excel export`), committed to `main` and pushed to `origin/main` on 2026-09-28.
- **Working Tree State:** Verified clean immediately after commit/push `6d28568` (zero unstaged, staged, or untracked modifications).
- **Last Explicitly Established Production Application Checkpoint:** Commit `7c89611` (`fix(security): enforce IT Admin intake boundary, capability-based visibility matrix, and regression suite`).
- **Production Deployment Status:** Deployment of `6d28568` to live hosting platforms (Railway backend / Vercel frontend) is `DEPLOYMENT UNVERIFIED / PRODUCTION UNVERIFIED`. Live state must be confirmed directly through cloud infrastructure dashboards.
- **Authoritative Database Migrations:** Repository migration directory is `backend/migrations/versions/`. Repository migration HEAD is `028_client_approval_workflow`. Production database revision was previously established at `028_client_approval_workflow`; current production state must not be assumed without independent verification before operational change.

2. Automated Test Execution Evidence

- **Release Gate Full Suite Result: 413 passed, 1 skipped** (414 total collected across 68 test files).
- **Skipped Test Verification:** `backend/tests/test_supabase_e2e.py::test_supabase_live_integration` was skipped because `TEST_POSTGRES_DB` was not set. The skip is environment-gated and does not represent a test failure.
- **Code Quality & Build Gates:** Python Ruff linter passed with zero errors across all modified backend files; Frontend ESLint passed with zero errors; Frontend production asset compilation (`npm run build`) completed in 24.69 seconds transforming 3,059 modules with zero errors.

3. Feature-by-Feature Technical Verification & UAT Matrix

Feature	Repo Status	Frontend Evidence	Backend Evidence	DB / Migration	Auth Dependency	Automated Test	Deployment	Live UAT Status	Client Reported Issue	Observed Defect / Gap	Final Classification	Next Verification Action
Auth / Password Login	IMPLEMENTED IN SOURCE	src/pages/Login.jsx	app/auth/router.py (POST /auth/login)	users, user_sessions	Public endpoint	test_auth.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	None observed in code	**LIVE VERIFICATION REQUIRED**	Execute live login test with synthetic account
MFA / OTP Verification	IMPLEMENTED IN SOURCE	src/components/auth/OtpVerifyDialog.jsx	app/auth/router.py (POST /auth/verify-otp)	users (otp_secret)	Pre-auth session	test_auth_otp.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	None observed in code	**LIVE VERIFICATION REQUIRED**	Execute live OTP challenge and verification flow

User Directory Admin	IMPLEMENTED IN SOURCE	src/pages/AdminDashboard.jsx	app/api/v1/users.py (GET /users)	users, user_roles, roles	ADMIN_USERS_MANAGE	test_users.py	UNVERIFIED (7c89611 est.)	BLOCKED	None reported	None observed in code	LIVE VERIFICATION REQUIRED	Verify admin roster display under authenticated session
User Excel Export	IMPLEMENTED & COMMITTED	src/pages/AdminDashboard.jsx (handleExport)	app/api/v1/users.py (GET /users/export)	users, audit_logs	ADMIN_USERS_MANAGE	test_user_excel_export.py (21 passed)	DEPLOYMENT UNVERIFIED (6d28568)	NOT DEPLOYED	None reported	None in code; deployment unconfirmed	IMPLEMENTED & COMMITTED	Verify live deployment of commit 6d28568; run export test
Canonical Person Identity	IMPLEMENTED IN SOURCE	src/pages/PersonDetail.jsx	app/api/v1/persons.py	persons, 027_person_numeric_id	person.view_sensitive	test_persons.py	UNVERIFIED (7c89611 est.)	BLOCKED	None reported	Non-gapless 10-digit ID; not citizen ID	LIVE VERIFICATION REQUIRED	Verify 10-digit Person ID generation in live environment
Person Duplicate Search	IMPLEMENTED IN SOURCE	src/components/person/PersonSearchModal.jsx	app/services/duplicate_service.py	persons	person.read	test_duplicate_service.py	UNVERIFIED (7c89611 est.)	BLOCKED	None reported	Search-before-create active; reduces duplicates	LIVE VERIFICATION REQUIRED	Execute live duplicate match test using synthetic data
Person Photos & Documents	IMPLEMENTED IN SOURCE	src/components/person/PersonPhoto.jsx	app/api/v1/documents.py	documents, 018_document_vault	document.read	test_documents.py	UNVERIFIED (7c89611 est.)	BLOCKED	None reported	HMAC-signed expiring URLs used	LIVE VERIFICATION REQUIRED	Test image retrieval and URL expiry behavior
Client Creation & Person Reuse	IMPLEMENTED IN SOURCE	src/pages/ClientNew.jsx	app/api/v1/clients.py (POST /clients)	clients, persons	client.create	test_clients.py	UNVERIFIED (7c89611 est.)	BLOCKED	None reported	Reuses existing Person	LIVE VERIFICATION REQUIRED	Verify client intake links to canonical Person
Client Approval Lifecycle	IMPLEMENTED IN SOURCE	src/pages/ClientDetail.jsx	app/api/v1/clients.py	028_client_approval_workflow	CLIENT_APPROVE	test_client_approvals.py	UNVERIFIED (7c89611 est.)	BLOCKED	None reported	Approval does not grant medical access	LIVE VERIFICATION REQUIRED	Execute propose -> approve/return/decline lifecycle
Client Medical Privacy	IMPLEMENTED IN SOURCE	src/components/client/MedicalProfileTab.jsx	app/api/v1/clients.py	client_medical_profiles	client.medical.read	test_client_medical.py	UNVERIFIED (7c89611 est.)	BLOCKED	None reported	Independent health capability enforced	LIVE VERIFICATION REQUIRED	Verify non-authorized user receives 403 on medical profile
Intake Reporting & Logging	IMPLEMENTED IN SOURCE	src/pages/IntakeNew.jsx	app/api/v1/referrals.py (POST /intake)	referrals, referral_reporters	referral.create	test_referrals.py	UNVERIFIED (7c89611 est.)	BLOCKED	None reported	None observed in code	LIVE VERIFICATION REQUIRED	Create synthetic intake concern and verify logging
Referral Screening	IMPLEMENTED IN SOURCE	src/pages/IntakeDetail.jsx	app/api/v1/referrals.py	intake_decisions, child_dispositions	referral.approve	test_intake_screening.py	UNVERIFIED (7c89611 est.)	BLOCKED	None reported	Transitions to Client/Case supported	LIVE VERIFICATION REQUIRED	Verify disposition screening transition to Case
Front Desk Inquiries & Routing	PARTIAL IN SOURCE	src/pages/FrontDeskDashboard.jsx	app/api/v1/front_desk.py	023_front_desk_foundation	front_desk.intake.view	test_front_desk.py	UNVERIFIED (7c89611 est.)	BLOCKED	Reported issues with inquiry queue	Missing durable owner, interaction notes, follow-up	REPORTED LIVE ISSUE	Investigate live Front Desk queue behavior and route handoff

Navigator Coordination	IMPLEMENTED IN SOURCE	src/pages/NavigatorDashboard.jsx	app/api/v1/referrals.py	referrals	navigator.referral.view	test_navigator.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	Isolated from open case management authority	**LIVE VERIFICATION REQUIRED**	Verify Navigator queue access limits
Case Management & Notes	IMPLEMENTED IN SOURCE	src/pages/CaseDetail.jsx	app/api/v1/cases.py, case_notes.py	cases, case_notes, case_restrictions	case.read_restricted	test_cases.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	Restricted case isolation active	**LIVE VERIFICATION REQUIRED**	Verify restricted case masking under standard caseworker
PlacementHome Registry	IMPLEMENTED IN SOURCE	src/pages/ResourceHomes.jsx	app/api/v1/resource_units.py	placement_homes, 015_resource_unit	resource_home.view	test_resource_units.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	Tracks capacity and home members	**LIVE VERIFICATION REQUIRED**	Verify alternate care home listing and capacity count
Caregiver Compliance & Visits	IMPLEMENTED IN SOURCE	src/pages/ResourceHomeDetail.jsx	app/api/v1/resource_units.py	placement_home_visits, licenses	resource_home.manage	test_resource_compliance.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	Visit logging supported; cadences not hardcoded	**LIVE VERIFICATION REQUIRED**	Verify visit log entry and license expiry display
Assistive Placement Matching	IMPLEMENTED IN SOURCE	src/pages/PlacementMatching.jsx	app/api/v1/placement_matching.py	placement_homes, clients	placement.match	test_placement_matching.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	Strictly assistive decision support (non-autonomous)	**LIVE VERIFICATION REQUIRED**	Verify matching scoring suggestions without auto-placement
Finance Service Requests	IMPLEMENTED IN SOURCE	src/pages/FinanceDashboard.jsx	app/api/v1/finance.py	service_requests, 011_finance_billing	finance.request.create	test_finance.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	Decimal/NUMERIC money; multi-step review	**LIVE VERIFICATION REQUIRED**	Test service request submission and approval workflow
Invoicing & Void/Reissue	IMPLEMENTED IN SOURCE	src/pages/FinanceInvoices.jsx	app/api/v1/finance.py	invoices, invoice_items	finance.invoice.manage	test_invoice_immutability.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	ADR-025 void/reissue semantics (no in-place edits)	**LIVE VERIFICATION REQUIRED**	Verify invoice voiding and reissue controls
HR Hub & Directory	IMPLEMENTED IN SOURCE	src/pages/HRDashboard.jsx	app/api/v1/org_ops.py (/org-ops)	employees (models/org_ops.py)	hr.employee.read	test_org_ops.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	Generic certification expiry tracking	**LIVE VERIFICATION REQUIRED**	Verify employee roster and certification alert display
CEO Command Centre	IMPLEMENTED IN SOURCE	src/pages/CEODashboard.jsx	app/api/v1/ceo.py	executive_initiatives, af1ad2d	executive.dashboard.view	test_ceo_dashboard.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	Aggregated operational metrics; no case records	**LIVE VERIFICATION REQUIRED**	Verify CEO dashboard metrics under executive account
Board Dashboard	IMPLEMENTED IN SOURCE	src/pages/BoardDashboard.jsx	app/api/v1/board.py	board_actions, 7df63f1	board.dashboard.view	test_board_dashboard.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	Board is not superuser; no client case access	**LIVE VERIFICATION REQUIRED**	Verify Board meeting publication and client data isolation
Document Vault	IMPLEMENTED IN SOURCE	src/components/shared/DocumentViewer.jsx	app/api/v1/documents.py	documents, document_access_events	document.read	test_documents.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	None reported	HMAC-signed expiring URLs; access audited	**LIVE VERIFICATION REQUIRED**	Test document upload, download token expiry, and audit log
Fleet Real-Time Checkout	PARTIAL IN SOURCE	src/pages/FleetManagement.jsx	app/api/v1/fleet.py	vehicles, vehicle_trips	fleet.trip.create	test_fleet.py	UNVERIFIED (7c89611 est.)	**BLOCKED**	Reported fleet workflow issues	Concurrency lock on active trip; advance booking missing	**REPORTED LIVE ISSUE**	Investigate vehicle checkout and double-booking conflict lock

Fleet Maintenance Tracking	IMPLEMENTED IN SOURCE	src/pages/FleetManagement.jsx	app/api/v1/fleet.py	vehicle_maintenance	fleet.vehicle.manage	test_fleet.py	UNVERIFIED (7c89611 est.)	BLOCKED	None reported	30-day maintenance reminder calculation	LIVE VERIFICATION REQUIRED	Verify scheduled maintenance reminders in live fleet panel
Fleet Advance Reservations	PARTIAL IN SOURCE	src/pages/FleetManagement.jsx	app/api/v1/fleet.py	vehicle_trips	fleet.trip.create	None (advance booking)	UNVERIFIED	BLOCKED	Reported inability to reserve vans ahead	Advance booking calendar and conflict engine absent	PARTIAL IN SOURCE	Design and build advance reservation calendar (Phase 3)
Fleet Fuel & Key Logs	NOT IMPLEMENTED	None	None	None	None	None	NOT APPLICABLE	NOT APPLICABLE	Reported missing fuel & key tracking	Identified in Office Coordinator discovery; unbuilt	CONFIRMED REQUIREMENT	Build FuelLog and key accountability workflow (Phase 3)
Physical Vehicle GPS	NOT IMPLEMENTED	None	TELEMATICS_SAMSARA (DISABLED)	vehicle_locations (stub)	None active	None (live telematics)	NOT DEPLOYED (DISABLED)	NOT APPLICABLE	Confirmed leadership requirement	No hardware installed; telematics disabled in code	CONFIRMED REQUIREMENT	Resolve governance policy, employee notice, and vendor selection
Microsoft 365 / Graph Sync	PARTIAL IN SOURCE (STUB)	None	FakeMicrosoftGraphService	integration_registry	integration.m365.manage	test_m365_stub.py	NOT DEPLOYED (STUB)	NOT APPLICABLE	None reported	Local stub implementation only; live tenant sync inactive	PARTIAL IN SOURCE (STUB)	Scope live Azure AD Single-Tenant registration and Graph API
Sage ERP Interoperability	PARTIAL IN SOURCE (EXPORT)	src/pages/FinanceDashboard.jsx	app/api/v1/finance.py	service_requests	finance.export	test_finance.py	UNVERIFIED	BLOCKED	None reported	Export only; no live API/database connector	PARTIAL IN SOURCE (EXPORT)	Verify CSV/Excel export compatibility with Sage General Ledger
Ask Red Bear AI Assistant	PARTIAL IN SOURCE (STUB)	src/components/ai/AskRedBearModal.jsx	app/services/ai.py (FakeAiProvider)	ai_request_audits	ai.ask_red_bear	test_ai_stub.py	NOT DEPLOYED (DISABLED)	NOT APPLICABLE	None reported	CRBCL_AI_ENABLED=False by default; fake provider only	PARTIAL IN SOURCE (DISABLED)	Requires leadership review before connecting live LLM
Reception Visitor Log	NOT IMPLEMENTED	None	None	None	FRONT_DESK_VISITOR_MANAGE	None	NOT APPLICABLE	NOT APPLICABLE	Front desk requested visitor sign-in	Permission constant exists; model/UI genuinely unbuilt	CONFIRMED REQUIREMENT	Build approved visitor-management model (Phase 2)
Visitation Room Booking	NOT IMPLEMENTED	None	None	None	None	None	NOT APPLICABLE	NOT APPLICABLE	Staff requested meeting room booking	CalendarEvent exists; structured room entity unbuilt	CONFIRMED REQUIREMENT	Build Room and RoomBooking models (Phase 2)
Public Website Intake Form	PARTIAL IN SOURCE (WEBHOOK)	None (external site)	POST /front-desk/ingest/google-form	public_intake_conversion_links	Public webhook	test_front_desk_ingest.py	UNVERIFIED	BLOCKED	None reported	Legacy Google Form webhook; DNS A-record boundary	PARTIAL IN SOURCE	Review public intake requirements and online form architecture

4. Technical Governance & Core Architectural Notes

A. Canonical Person Identifier: Each Person receives an automatically generated 10-digit numeric CRBCL Person ID. The current implementation uses sequence base `1100000000` with database row locking introduced by migration `027_person_numeric_id`. The sequence base is an implementation choice, not CRBCL policy or a citizenship identifier, and no gapless-sequence guarantee is asserted.

B. User Export Architecture: The bulk User Excel export (commit `6d28568`) provides administrative user auditing capabilities under the following technical design:
• Export Structure: Generates a six-column workbook: Username, First Name, Last Name, Status, Email, and Last Login.
• Data Mapping Limitation: The underlying `users` database schema stores `email` and `full_name`. In the export mapping, Username maps directly to `User.email`; First Name and Last Name are parsed dynamically from `User.full_name` and do not represent independently authoritative database fields.
• Authoritative Last Login Behavior: Last Login is sourced exclusively from `User.last_login_at`. Successful password and OTP authentication update it through consolidated successful-login bookkeeping; failed authentication and token refresh do not. Users who have never successfully logged in export a blank value.
• Audit Trail: Every directory export operation records a `USER_DIRECTORY_EXPORTED` event in `audit_logs` capturing `user_id`, timestamp, `row_count`, `export_format`, `filename`, and client IP address (`ip_address`).

C. Role Segregation & Administrative Boundaries: Platform authorization enforces explicit capability boundaries:
• Board Membership Boundary: Board membership does not make a user a superuser or grant Person, Client, or Case access.
• IT Administration Boundary: IT administration does not inherently grant access to protected Client records, Intake narratives, HR dossiers, or protected governance content; each protected domain requires its applicable authorization. Access to sensitive child health and family records requires explicit, independent capability grants (`client.medical.read`).

5. External Integrations Reality Assessment

- **1. Microsoft 365 / Exchange / Teams:** Live Microsoft 365 / Graph integration is not active in the platform; the repository uses local stub implementations (`FakeMicrosoftGraphService`).
- **2. Sage ERP (Finance):** No live Sage API or database connector has been established. Interoperability is currently through exports where supported; the report does not characterize unsupported workflows as integrated.
- **3. Ask Red Bear (AI Assistant):** Backed by `FakeAiProvider`. AI capabilities are disabled by default via configuration (`CRBCL_AI_ENABLED=False`); no commercial LLM endpoint is connected.
- **4. Fleet Telematics (Samsara / GPS):** Physical tracking devices are not deployed. The integration registry keeps telematics disabled; no physical tracker or live telematics integration is currently implemented.
- **5. Public Intake / WordPress:** The public website is hosted externally. A DNS A record maps hostname to IPv4; it does not transmit form submissions. Google Form webhook ingestion exists in code but is not active or approved.

6. Architectural Boundaries & Security Controls

- **Access Control:** Enforced via application service logic, FastAPI authorization dependencies, and capability matrices (`app/core/auth.py`), ensuring strict boundaries between administrative and child welfare roles.
- **Transport & Document Access Security:** Public application traffic uses HTTPS/TLS where deployed accordingly. Protected document delivery uses signed, expiring download URLs and document release controls.
- **Audit Trail:** Application audit records capture instrumented security and operational events, including administrative exports and protected operations. Where histories are described as append-only, that qualification refers to verified application behavior unless database-level immutability is separately established.

7. Live Verification Limitations & Blocker Analysis

During this forensic re-audit, end-to-end live testing was blocked due to the following specific constraints:
1. **Credential Boundary:** No live production administrative credentials were provided for automated or browser-based testing.
2. **Network Ingress Boundary:** Direct external HTTP probes to reported live endpoints (`crbcl-production.up.railway.app`, `genserver.online`) timed out, preventing automated endpoint validation.
3. **Data Governance Boundary:** In strict adherence to CRBCL data governance rules, no live community records may be inspected, created, or mutated.

Conclusion: In accordance with reporting rules, all capabilities with source implementation but unverified live runtime are classified as `LIVE VERIFICATION REQUIRED / BLOCKED`. Capabilities with specific frontline defect reports are classified as `REPORTED LIVE ISSUE — INVESTIGATION REQUIRED`.

8. Reporting and Assurance Boundaries

- **1. Statutory and Regulatory Alignment:** No assertion of FOIP, LA FOIP, HIPA, or OCAP compliance or certification is made. The platform provides supporting technical controls; formal statutory compliance and applicability require CRBCL legal and privacy counsel review.
- **2. Production Deployment Baseline:** Commit `6d28568` is current repository HEAD and is `PRODUCTION UNVERIFIED`; production application status is not asserted from Git history alone.
- **3. Visitor Management Status:** A visitor log is not implemented in the current codebase; it is documented as a requested capability from administrative discovery.

- **4. Fleet Scheduling Scope:** Advance vehicle reservation calendars, key accountability logs, and recurring booking conflict resolution are not implemented; current functionality is restricted to real-time checkout conflict locking.
- **5. Vehicle Telematics Status:** Physical GPS telemetry and manual coordinate capture workflows are not implemented; no physical tracking hardware is active.
- **6. Caregiver Payroll and Fiscal Reconciliation:** Autonomous caregiver payroll and automated Indigenous Services Canada (ISC) funding reconciliation are not implemented; financial capabilities support rate versioning and purchase order workflows.
- **7. Operational Policy Cadence and Thresholds:** No operational policy cadences (annual licensing, quarterly monitoring), retention schedules, response SLAs, or financial approval dollar thresholds are asserted as established CRBCL policy.
- **8. Person Identifier Scope:** The 10-digit numeric sequence (base `1100000000`) is an internal technical identifier and is not represented as a citizenship registry or statutory identity document.
- **9. Operational Readiness Boundary:** Controlled UAT readiness does not constitute production release approval or authorization for live community data operations.